



MUTUAL FUND ANALYTICS PROJECT

Bluestock Fintech Capstone Project

Prepared By:

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Tools Used:

Python | SQL | SQLite | PostgreSQL | Power BI | Pandas |
Numpy | Matplotlib | Seaborn | Scipy | Jupyter Lab | VS code

Project Duration: 7 Days

Submission Date: June 2026

Executive summary

This project focuses on analyzing mutual fund performance, investor transactions, portfolio holdings, and NAV history using Python, SQL, and Power BI.

The objective of the project is to build a complete data analytics pipeline that collects, cleans, stores, analyzes, and visualizes mutual fund data for business insights and investment decision support.

The project includes:

- * Data ingestion from multiple mutual fund datasets
- * Data cleaning and validation
- * SQL database design and implementation
- * Exploratory Data Analysis (EDA)
- * Performance and risk analysis
- * Investor behavior analysis
- * Dashboard creation using Power BI
- * Recommendation engine for mutual fund selection

The final dashboard provides insights into fund performance, risk metrics, investor demographics, SIP behavior, benchmark comparison, and portfolio diversification.

This project demonstrates practical skills in data engineering, data analysis, business intelligence, and financial analytics.

DATA SOURCES

The Mutual Fund Analytics project uses multiple datasets covering fund information, NAV history, investor transactions, scheme performance, and portfolio holdings. These datasets were used to perform data cleaning, exploratory analysis, performance evaluation, risk assessment, and dashboard development.

Dataset Overview

1. Fund Master Data

- Contains scheme information such as AMFI code, fund house, category, sub-category, and scheme name.
- Used as the primary reference table across the project.

2. NAV History Data

- Contains historical Net Asset Value (NAV) records for mutual fund schemes.
- Used for return calculations, trend analysis, volatility measurement, and risk metrics.

3. Investor Transaction Data

- Includes SIP, redemption, and lump-sum transactions.
- Used for investor behavior analysis and SIP continuity analysis.

4. Scheme Performance Data

- Contains fund performance metrics such as returns, Sharpe ratio, expense ratio, AUM, and risk grade.
- Used for fund ranking and recommendation analysis.

5. Portfolio Holdings Data

- Contains stock holdings and sector allocation information.
- Used for concentration analysis and diversification assessment.

Data Source Summary

Dataset | Purpose

Fund Master | Scheme information

NAV History | Return and risk analysis

Investor Transactions | Investor behavior analysis

Scheme Performance | Fund performance evaluation

Portfolio Holdings | Portfolio concentration analysis

The datasets were cleaned, validated, and transformed before being loaded into the analytics pipeline and SQL database.

Project Architecture & ETL Design

The project follows a complete ETL (Extract, Transform, Load) pipeline to process mutual fund datasets and generate actionable business insights.

ETL Workflow

1. Extract Phase

Data was collected from multiple CSV datasets and mutual fund NAV APIs. The extracted datasets included:

- Fund Master Data
- NAV History Data
- Investor Transactions
- Scheme Performance Data
- Portfolio Holdings Data

2. Transform Phase

The extracted data was cleaned and validated before analysis

Data transformation activities included:

- Handling missing values
- Removing duplicate records
- Standardizing date formats
- Validating numerical columns

- Calculating daily returns
- Computing rolling performance metrics
- Creating risk and investor analysis datasets

3. Load Phase

The cleaned datasets were loaded into a SQLite database for structured storage and querying.

Database tables included:

- dim_fund
- fact_nav
- fact_transactions
- fact_performance
- fact_portfolio

Project Architecture

Data Sources (CSV Files & APIs)



Python ETL Scripts



Data Cleaning & Validation



SQLite Database



EDA & Advanced Analytics



Power BI Dashboard



Business Insights & Recommendations

Technologies Used

- Python
- Pandas
- NumPy
- SQLite
- SQL
- Matplotlib
- Seaborn
- Power BI
- VS Code
- Git & GitHub

The ETL pipeline ensures data consistency, quality, and reliability for analytics and dashboard reporting.

EDA FINDINGS – Part 1

The Exploratory Data Analysis (EDA) phase was performed to understand the structure, quality, and distribution of the mutual fund datasets. The objective was to identify patterns, trends, anomalies, and business insights before performing advanced analytics.

Dataset Overview

The project utilized multiple datasets containing mutual fund information, historical NAV values, investor transactions, and portfolio holdings. Initial analysis was conducted to understand:

- Number of records and columns
- Missing values
- Duplicate records
- Data types
- Distribution of key variables

Key EDA Activities

1. Missing Value Analysis

Missing values were identified and handled appropriately using data cleaning techniques. Null values in critical fields were either imputed or removed based on business requirements.

2. Duplicate Record Analysis

Duplicate records were checked across all datasets to ensure data consistency and improve analytical accuracy.

3. Fund Category Distribution

Analysis of fund categories revealed the distribution of schemes across different investment segments such as:

- Large Cap Funds
- Mid Cap Funds
- Small Cap Funds
- Hybrid Funds
- Debt Funds

This helped understand the composition of the mutual fund universe.

4. NAV Trend Analysis

Historical NAV data was analyzed to observe:

- Long-term growth patterns
- Market corrections
- Volatility periods
- Performance consistency

Funds with strong NAV growth generally demonstrated better long-term performance characteristics.

Key Findings

- Most schemes showed positive long-term NAV growth.
- Large-cap funds exhibited lower volatility compared to mid-cap and small-cap funds.
- Data quality after cleaning was suitable for advanced analytics and dashboard development.
- Historical NAV trends highlighted the impact of market cycles on mutual fund performance.

EDA FINDINGS (Part – 2)

Investor Transaction Analysis

Investor transaction data was analyzed to understand investment behavior and transaction patterns across different mutual fund schemes.

Key observations included:

- SIP transactions contributed a significant portion of total investments.
- Transaction activity varied across schemes and categories.
- Long-term investors showed higher investment consistency.

Portfolio Holdings Analysis

Portfolio holdings data was examined to understand asset allocation and concentration levels.

The analysis focused on:

- Sector-wise exposure
- Top stock holdings
- Portfolio diversification
- Concentration risk

Scheme Performance Distribution

Scheme performance metrics such as returns, risk grade, Sharpe ratio, and expense ratio were analyzed.

Observations:

- High-return schemes generally carried higher risk.
- Risk-adjusted performance varied significantly across schemes.
- Certain schemes demonstrated consistent long-term performance.

EDA Summary

The exploratory analysis helped identify important trends and relationships within the data. These findings served as the foundation for advanced analytics, risk assessment, recommendation modeling, and dashboard development.

Performance Analysis

Performance analysis was conducted to evaluate the effectiveness of mutual fund schemes using key financial metrics.

Metrics Evaluated

- Annual Returns
- Sharpe Ratio
- Expense Ratio
- Risk Grade
- Assets Under Management (AUM)

Findings

- Funds with higher returns generally exhibited higher volatility.
- Some schemes delivered strong risk-adjusted performance despite moderate returns.
- Expense ratios influenced long-term investor outcomes.
- Large-cap funds showed relatively stable performance compared to higher-risk categories.

Performance metrics were used to rank schemes and support investment recommendations.

Advanced Analytics

Advanced analytical techniques were applied to evaluate mutual fund risk, performance stability, and investor behavior.

Rolling Sharpe Ratio

Rolling Sharpe Ratio was calculated to measure risk-adjusted returns over time. It helped identify periods of strong and weak performance across different market conditions.

Value at Risk (VaR)

Value at Risk (VaR) was used to estimate the maximum potential loss at a given confidence level. This metric helped assess downside risk exposure.

Conditional Value at Risk (CVaR)

CVaR was calculated to understand the average loss beyond the VaR threshold. It provided deeper insights into extreme market risk scenarios.

HHI Concentration Analysis

The Herfindahl-Hirschman Index (HHI) was used to evaluate portfolio concentration. Lower HHI values indicated better diversification, while higher values suggested concentrated portfolios.

SIP Continuity Analysis

Investor transaction data was analyzed to identify SIP continuation patterns and long-term investment behavior.

Power BI Dashboard Overview

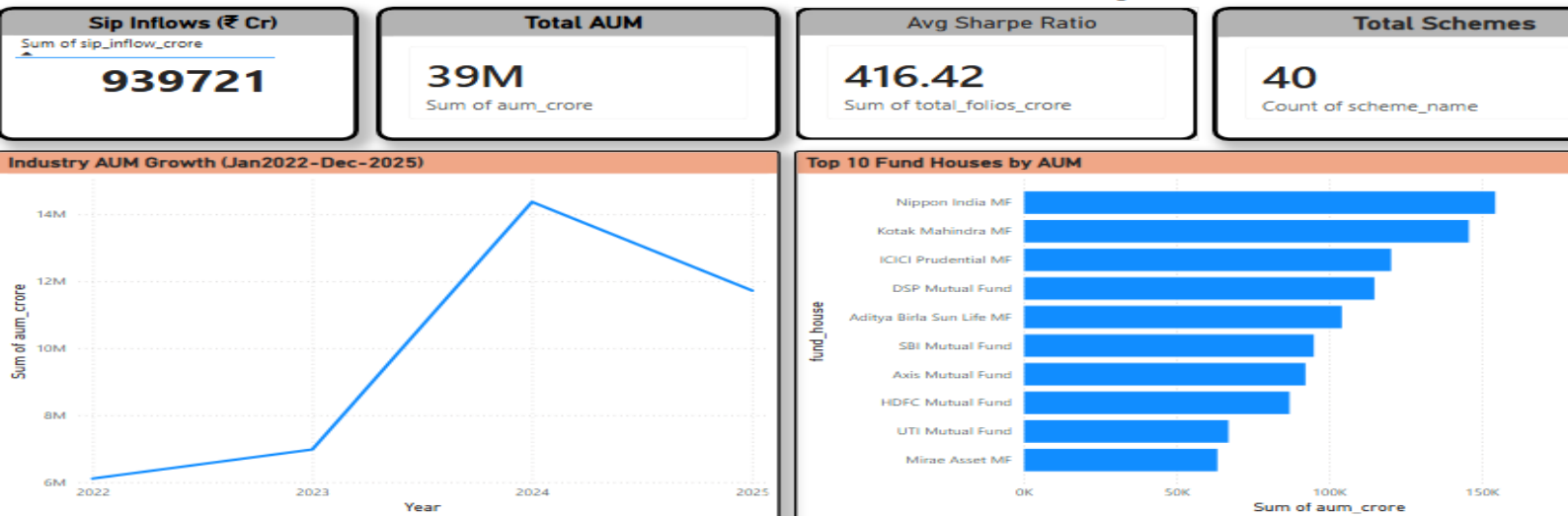
An interactive Power BI dashboard was developed to visualize key metrics and insights generated throughout the project.

Dashboard features included:

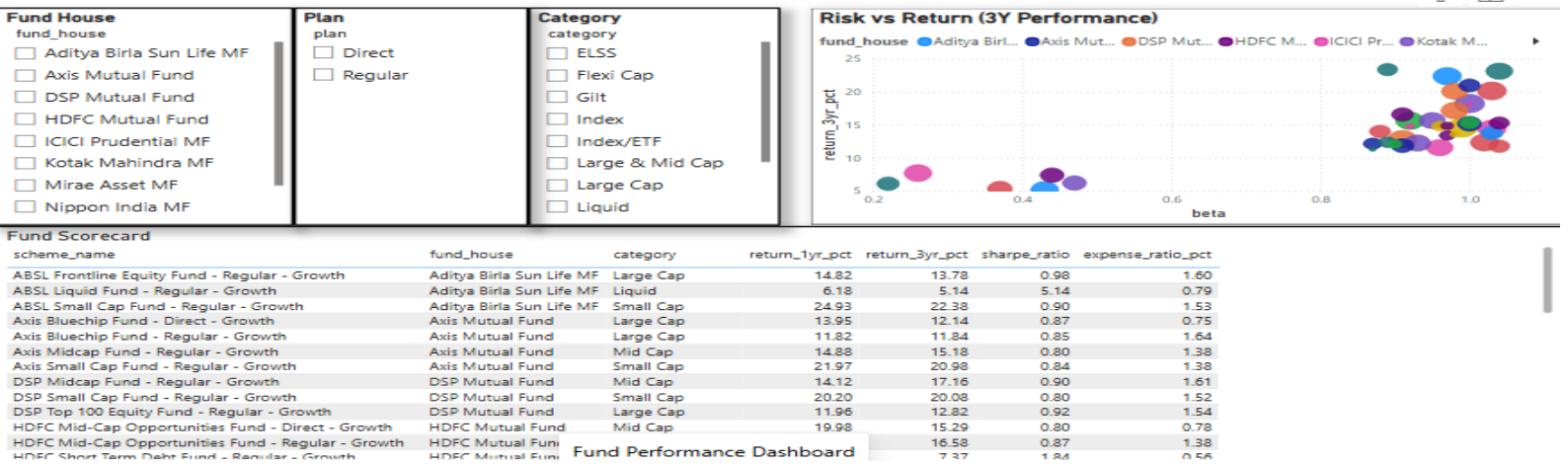
- SIP vs NIFTY Benchmark Comparison
- Investor Transaction Analysis
- Fund Category Distribution
- NAV Trend Analysis
- Portfolio Holdings Overview
- Risk and Performance Metrics
- Investor Cohort Analysis

The dashboard enables users to interactively explore data, compare schemes, and identify investment trends through filters and visualizations.

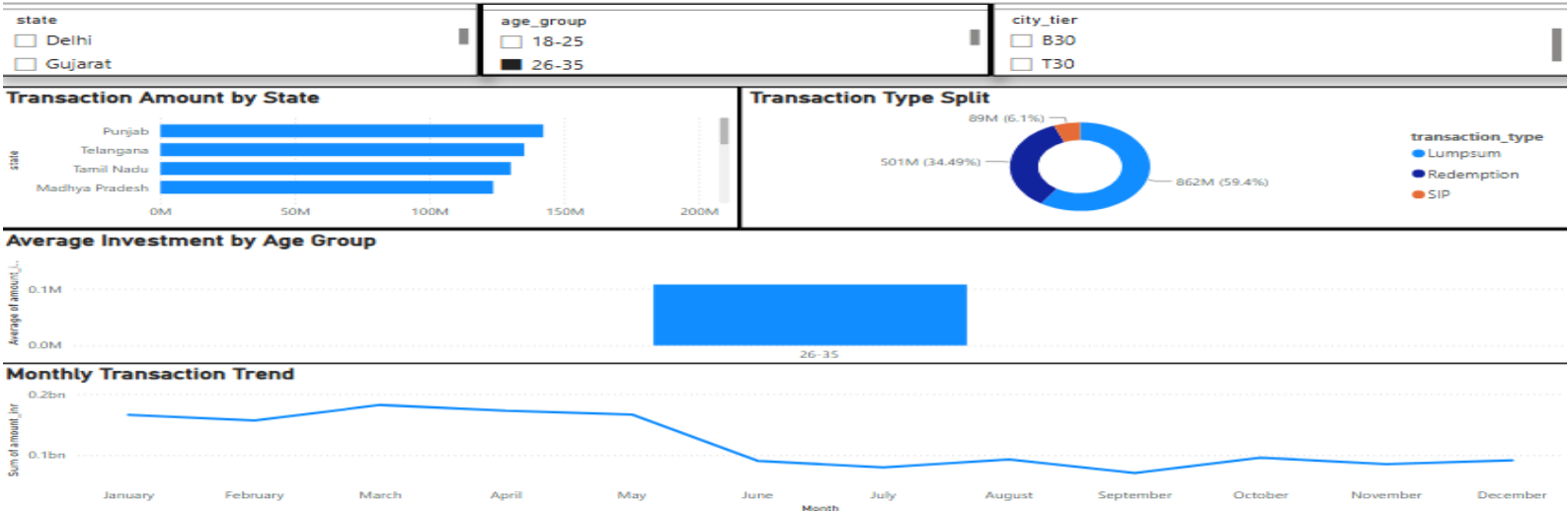
Mutual Fund Analytics Dashboard



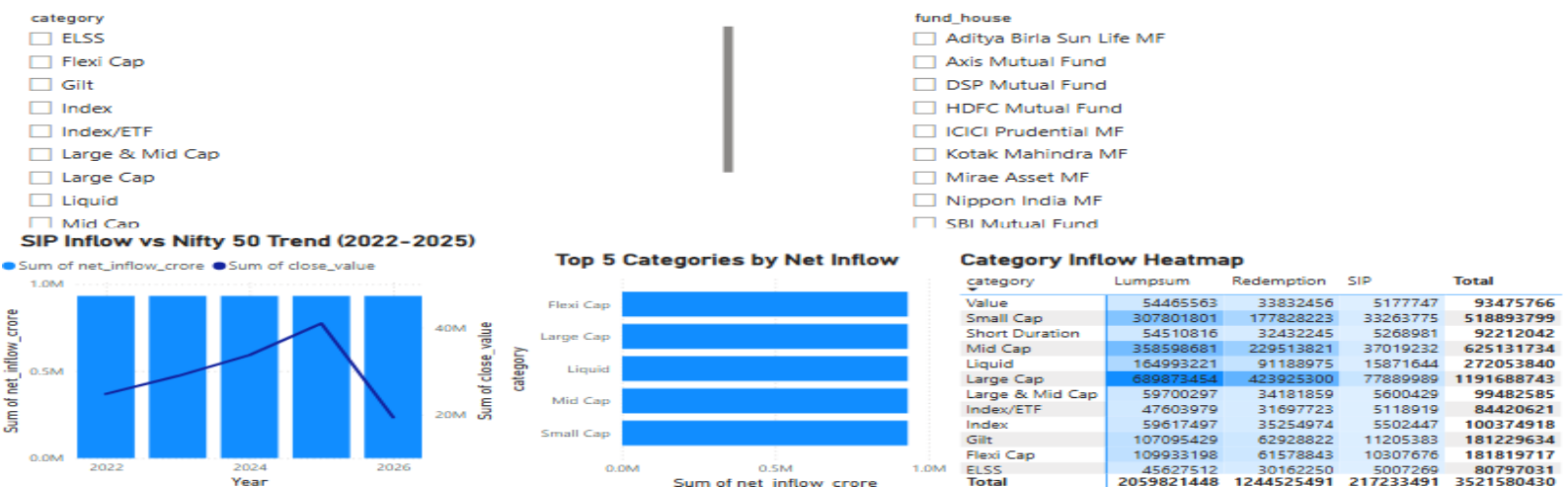
Fund Performance Dashboard



Investor Analytics Dashboard



SIP & Market Trends Dashboard



Key Findings

- Large-cap mutual funds demonstrated relatively stable long-term performance.
- SIP investments contributed significantly to investor wealth creation.
- Historical NAV analysis showed clear market cycle impacts.
- Risk-adjusted returns varied considerably across mutual fund schemes.
- Diversified portfolios exhibited lower concentration risk.
- Investor retention was stronger among consistently performing schemes.
- Advanced risk metrics helped identify downside risk exposure.
- Data-driven analysis improved fund comparison and investment evaluation.

Recommendations

Based on the analysis conducted, the following recommendations are proposed:

1. Encourage long-term SIP investments to maximize wealth creation.

2. Promote diversified portfolios to reduce concentration risk.
3. Monitor risk-adjusted performance metrics regularly.
4. Compare mutual fund performance against benchmark indices.
5. Improve investor awareness regarding risk and return trade-offs.
6. Utilize dashboard insights for data-driven investment decisions.
7. Review portfolio allocation periodically to maintain diversification.

Conclusion

This project successfully developed a complete Mutual Fund Analytics solution using Python, SQL, SQLite, and Power BI. The project covered data ingestion, cleaning, transformation, exploratory analysis, advanced analytics, dashboard development, and reporting.

The implemented analytics framework provides valuable insights into mutual fund performance, investor behavior, portfolio concentration, and risk assessment. The Power BI dashboard offers an intuitive interface for exploring and understanding mutual fund data.

Future Scope

Future enhancements may include:

- Real-time NAV integration
- Machine learning-based fund recommendation models
- Cloud deployment of analytics pipelines
- Automated reporting systems
- Mobile-friendly dashboard development

The project demonstrates practical applications of data analytics and business intelligence in the financial services domain